Chapter 1
INTRODUCTION

Oral health remains a persistent and measurable public health challenge among Filipino school children. The Department of Health's (DOH) 2021 National Monitoring and Epidemiological Dental Survey reported that 92% of Filipinos have dental caries, with 78% of 12-year-old children affected and an average Decayed, Missing, and Filled Teeth (DMFT) score of 4.48, exceeding the World Health Organization's standard of 3 DMFT [1], [2]. In response, the DOH has identified oral health as one of the eight priority areas under the National Objectives for Health (NOH) Philippines 2023–2028 and has mandated school dental clinics to provide regular screenings and preventive treatments such as bi-annual fluoride varnish application through its School Dental Health Care Program [3], [4].

Table 1.1. *Summary of Selected Dental Health Statistics/Indicators*

Indicator/Statistic	Value	Population/Scope
Prevalence of dental caries	92%	Filipinos
% of 12-year-old children affected	78%	Filipino school children
Standard DMFT score	3	Global
Average DMFT score	4.48	Filipino school children

Despite this mandate, school dental clinics, including those in Barangay Tanyag, Taguig City, continue to rely on manual, paper-based processes for managing student records, scheduling appointments, and generating reports. These practices introduce significant inefficiencies and the resulting fragmentation of data limits the clinics' ability to monitor student dental health trends, ensure follow-up care, and make timely, informed clinical decisions.

Prior studies have demonstrated that electronic health records (EHRs) in dental practice can significantly reduce missing information and improve coordination among healthcare providers [5], [6]. Research has further shown that predictive analytics applied to dental records can support early intervention by identifying students at higher risk of developing oral health conditions and enabling more proactive treatment planning [7], [8], [9]. However, a gap persists between these demonstrated capabilities and actual implementation in resource-constrained, barangay-level school dental settings, where processes remain largely manual and fragmented.

This study addresses that gap by proposing the design and development of Floral, a Dental Health Record Management System that centralizes student dental and treatment records across schools, enhances appointment management, and supports data-driven decision-making through automated reporting, with integrated predictive analytics in Barangay Tanyag, Taguig City to identify potential dental health risks and facilitate preventive dental care planning among students.

Background of the Study

The urgency of addressing school dental health is further reinforced by its alignment with the United Nations Sustainable Development Goals (SDGs). Most directly, SDG 3: Good Health and Well-Being seeks to ensure health and well-being for all at every stage of life, including access to quality and affordable healthcare services. Improving access to dental care for school children through a digitized and efficient system directly contributes to this goal [10]. The study also aligns with SDG 4: Quality Education, which recognizes that good health is a prerequisite for effective learning, as children who suffer from untreated tooth decay often experience pain and discomfort that can affect their attendance, concentration, and overall academic performance [11]. It further connects to SDG 10: Reduced Inequalities, as reducing inequality requires ensuring equal opportunity and inclusive social growth, particularly for young people and vulnerable communities with limited access to private dental care [12]. On an institutional level, the study aligns with SDG 16: Peace, Justice, and Strong Institutions, specifically Target 16.6, which calls for effective, accountable, and transparent institutions at all levels [13], promoted through structured recordkeeping, automated reporting, and an audit trail. Together, these goals highlight that improving school dental health record management is not only a local necessity but also a contribution to broader national and global health and development commitments.

Bridging these global commitments to the local level, however, reveals a contrast between policy intent and operational reality. In Barangay Tanyag, Taguig City, dental health services for students are delivered through three school dental clinics: Bagong Tanyag Integrated School, which serves as the dentist's main clinic and accommodates students from Kindergarten to Grade 10; South Daanghari Elementary School; and Bagong Tanyag Elementary School, both of which cover Kindergarten to Grade 6. Each year level has over eight to eleven sections, with each section comprising approximately 45 students, placing the total student population under the program's coverage at around 8,000 students. Despite this scale, the existing workforce consists of only one dentist, one dental aide, and three dental staff responsible for all three clinics. The clinic provides a range of dental services, including oral examinations, oral prophylaxis, fluoride varnish, pit and fissure sealant, silver diamine fluoride, and simple extractions. Rather than functioning as efficient, data-driven points of care, however, these clinics continue to operate under manual, paper-based systems that undermine the very goals these programs seek to achieve. Student records such as consent forms, dental health histories, and dental charts are maintained only on paper, making them vulnerable to loss or damage. Manual dental charting can be slow and prone to errors, while inconsistent handwriting and manual record updates may lead to misinterpretation of treatment records and delays in patient management. Scheduling and communication with parents are handled through paper forms, phone calls, or text messages, and the preparation of reports on the number of treated students and performed procedures is done manually, making data compilation time-consuming and prone to inaccuracies. Furthermore, while students are mandated to receive dental services twice a year [4], the clinic struggles to systematically monitor and complete these required visits. The sheer volume of approximately 8,000 student records, combined with limited personnel and persistent reliance on manual processes, further aggravates inefficiencies in record tracking, service delivery, and compliance with the twice-yearly mandate.

As a result, there is an increasing need to adopt digital health solutions, such as EHRs, to improve data management and service delivery in school dental clinics, which the DOH has also encouraged as part of its NOH agenda [3]. This shift toward digitization is no longer optional but a necessary step in modernizing school dental care delivery. Studies have confirmed that EHRs in dental practice significantly enhance clinical efficiency, reducing the risk of missing information and improving coordination among healthcare professionals [5][6]. Beyond digital record management, the integration of predictive analytics in healthcare systems has shown strong potential in supporting data-driven decision-making. Predictive models can identify students who are at higher risk of developing dental health issues, prioritize treatment needs, and assist in planning preventive interventions such as fluoride application, particularly beneficial in school dental clinics where resources and time are limited. Research has demonstrated that predictive analytics using dental records can support treatment planning and early intervention, with models showing strong capability in predicting untreated dental caries and enabling timely scheduling of preventive care [7][8][9]. The oral conditions recorded in each student's Individual Patient Treatment Record across successive clinical visits constitute the dental historical records that form the basis for this risk classification process. Despite this, a gap still exists between these technological advancements and their actual implementation in school dental clinics, where processes remain manual and technologies are not yet fully utilized.

These operational realities underscore the need for a digitized, integrated solution that can support the clinic's capacity to deliver consistent and data-driven dental care at the scale and frequency that students require. Addressing this need is the central motivation of this study.

Statement of the Objectives

General Objective

To design, develop, and evaluate Floral, a Dental Health Record Management System that centralizes student dental and treatment records across schools, enhances appointment management, and supports data-driven decision-making through automated reporting in Barangay Tanyag, Taguig City, with integrated predictive analytics to identify potential dental health risks and facilitate preventive dental care planning among students.

Specific Objectives

The present study will seek answers to the following objectives:

1. To develop a system that improves the management of student dental health records at Barangay Tanyag school dental clinics through Optical Character Recognition (OCR) covering the student personal information such as name, birthday, age, sex, address, contact number, grade level and section.

2. To design and develop a system that enables appointment scheduling and monitoring, logging of dental charts, recording and updating of individual patient treatment records (IPTR), routine preventive care (RPC) records, and centralized storage of student dental records across schools.
3. To integrate a predictive analytics module to analyze historical dental records and dental health conditions to predict risk classification interface through risk stratification categorization of dental health data and treatment recommendations to the dentist as inputs for a decision-support system.
4. To generate dashboards that provide an overview of student dental health, reports that summarize treatment history and appointment data, and filterable and searchable dental records that allow school clinic staff to monitor progress and manage follow-ups efficiently.
5. To test and evaluate the system using the ISO 25010, version 2023 standard on software product quality in terms of:
    5.1. Functional Suitability
    5.2. Performance Efficiency
    5.3. Compatibility
    5.4. Usability
    5.5. Reliability
    5.6. Security

Conceptual Framework

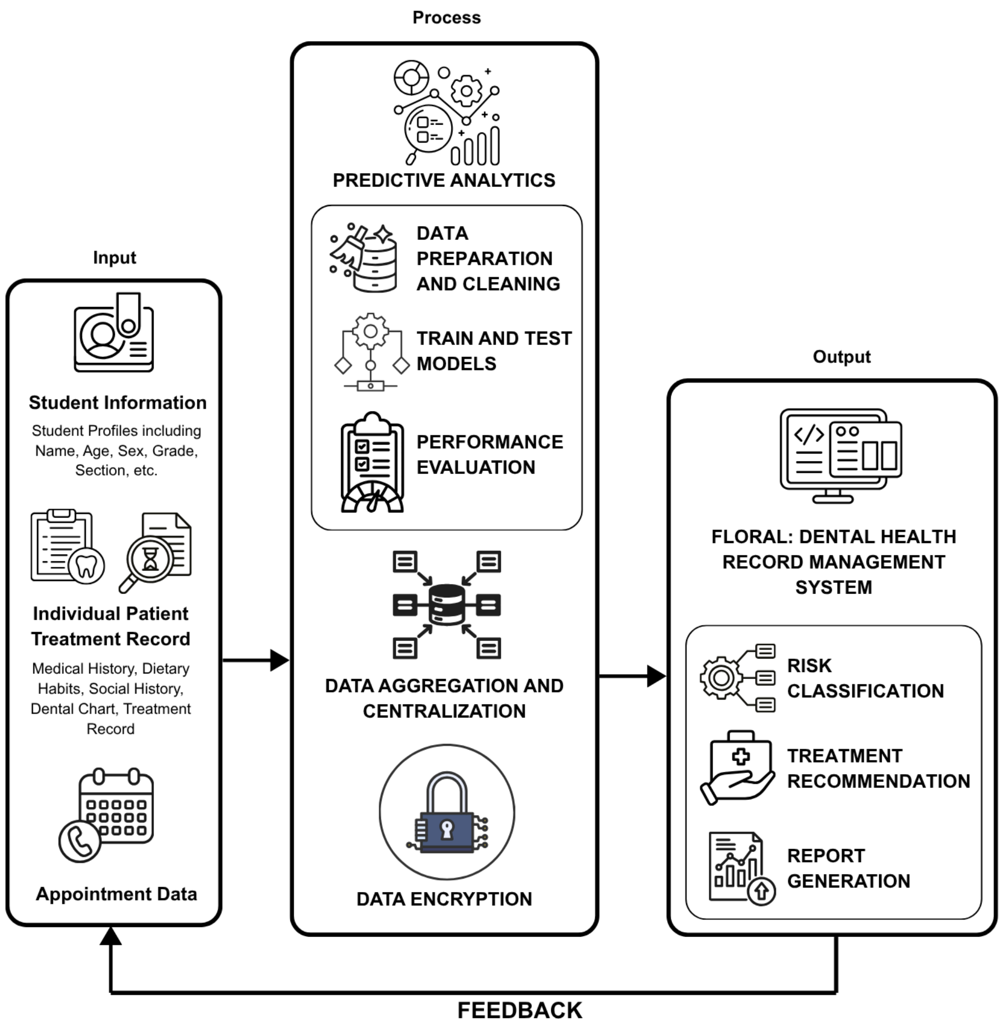

Figure 1.1. *Conceptual Framework*

The conceptual framework of this study is based on the Input-Process-Output (IPO) Model [14] , which systematically represents the flow of data and processes within the proposed Dental Health Record Management System. The framework is designed to centralize student dental records, enhance appointment management, and support data-driven decision-making through automated reporting and predictive analytics, ultimately facilitating preventive dental care planning across the three covered schools in Barangay Tanyag.

Input

The input component consists of all data required for system operation and analysis. Student information such as name, age, sex, grade level, and section establishes the foundation of each student's profile. The IPTR captures comprehensive medical history, dietary habits, social history, dental charting details, and treatment records. Appointment data is systematically recorded to manage schedules, follow-ups, and visit history. Together, these inputs provide the necessary data for the system to process, analyze, and generate meaningful outputs.

Process

The process component represents the core system operations that transform input data into meaningful outputs. The system begins with secure user authentication, implementing role-based access control for dentists, dental aides, and school administrators. Dental record management centralizes the recording and updating of student records across all covered schools, while appointment scheduling and monitoring manages clinic appointments, tracks visit status, and flags pending follow-ups. Dental charting functionality enables dentists to record tooth-by-tooth conditions using standard dental marking conventions. The RPC tracking module monitors the two-visit preventive care process per student.

The predictive analytics component applies machine learning to identify dental health risks and support clinical decision-making. Data preparation and cleaning is first performed to remove errors, handle missing values, and standardize data formats, ensuring the reliability of subsequent analysis. The system then performs data aggregation and centralization, combining student records from all three covered schools into a unified database for comprehensive analysis. Data encryption is applied to protect sensitive student health information, ensuring compliance with data privacy regulations.

The system proceeds to train and test models using machine learning algorithms to identify students' dental health issues. Performance evaluation is conducted using metrics to validate model reliability. Based on the predicted output, the system generates appropriate treatment recommendations as a decision-support tool for dentist review and validation.

Output

The output component delivers comprehensive results and insights generated by the system to support clinical decision-making and health monitoring. Centralized student dental records are unified, searchable, and filterable across all three schools. Interactive digital dental charts offer a visual, tooth-by-tooth representation of each student's dental health status using standard dental markings. RPC completion records document the two-visit preventive care progress per student, ensuring accountability and continuity of care. Risk classification results label students as High, Medium, or Low risk, allowing for targeted interventions, while treatment plan suggestions provide condition-appropriate recommendations for dentist review and validation. Appointment summaries and follow-up alerts ensure structured scheduling outcomes and timely reminders. Standardized monthly reports aggregate and structure data in alignment with DOH reporting standards, including providing age-bracket and gender-based counts for monitoring and decision-making [4]. The health dashboard provides a visual, school-wide overview of dental health trends, empowering administrators and public health officials with the insights needed to improve dental health outcomes across the community.

Significance of the Study

The present study may be beneficial to the following:

The Dentists. From the results of this study, they may experience a significant improvement in the efficiency and accuracy of dental record management through the use of a centralized digital system. The availability of interactive dental charts, automated patient records, and predictive-based treatment prioritization may enable them to make more informed clinical decisions, reduce the time spent on administrative tasks, and focus more on delivering quality dental care to students.

The Dental Aides. From the results of this study, dental aides may benefit from a more organized and streamlined workflow in supporting the delivery of dental services. The system may reduce the burden of manual recordkeeping and paperwork, allowing them to assist dentists more effectively and ensure that patient information is accurately recorded, updated, and readily accessible when needed.

The Students. From the results of this study, students may receive more consistent, timely, and equitable dental care. The system's ability to identify and prioritize students with the most urgent dental health needs may help ensure that no case is overlooked, potentially leading to earlier intervention, fewer complications, and improved dental health outcomes over time.

The School Dental Clinics. From the results of this study, school dental clinics as institutions may benefit from an efficient and standardized approach to managing dental health. The system may improve the overall capacity of school dental clinics to deliver timely dental care, supporting their mandate under the Department of Health's School Dental Health Care Program.

The School Administrators. From the results of this study, school administrators may benefit from a more structured and efficient approach to managing dental health data within their institutions. Automated report generation and organized record keeping may reduce their administrative workload.

The Barangay Health Center. From the results of this study, the Barangay Health Center may directly benefit from the use of the developed system since it follows the same dental processes, recordkeeping practices, and reporting requirements as the school dental clinics. The only difference lies in the age group they serve, as barangay health centers cater to infants and adolescents rather than school-age students. With minimal adjustments, the system may be adopted by the Barangay Health Center to improve the efficiency and accuracy of its own dental health services.

The City Health Office. From the results of this study, the Barangay Health Office may gain access to accurate, consolidated, and up-to-date data on the dental health status of students across all three covered schools. This may support more timely and evidence-based decision-making in the allocation of dental resources and the submission of required reports to the city health office.

The Community and Public Health. From the results of this study, the broader community may benefit from an improved model of school-based dental health service delivery at the barangay level. The system may demonstrate that meaningful advancements in preventive care and health data management are achievable even in resource-limited settings, potentially serving as a replicable framework for other barangay-level school dental programs across the Philippines.

The Researchers. This study may provide the researchers with a deeper understanding of the challenges and opportunities involved in developing and implementing a digital health system in a community-based school setting. The process of designing, developing, and evaluating the system may also strengthen their technical competencies and contribute to their growth as researchers in the field of health informatics and software development.

The Future Researchers. Based on the results of this study, future researchers may use the documented design, development process, and evaluation outcomes as a reference for conducting related studies in similar settings. This study may also serve as a foundation for further exploration of predictive analytics in school health programs and the development of more advanced digital health systems that build upon the findings presented in this research.

Scope and Limitations of the Study

Scope

This study focuses on the design and development of a web-based Dental Health Record Management System with integrated predictive analytics for the school dental clinics of Barangay Tanyag in Taguig City, specifically covering Bagong Tanyag Integrated School, Bagong Tanyag Elementary School Annex A, and South Daang Hari Elementary School Main. The subjects of the study include the students enrolled in these three schools, along with the dentists, dental aides, clinic staff, school administrators, and city health office who interact with the dental clinic. The system will cover the design and development of an appointment scheduling and monitoring, recording and updating of individual patient treatment records, logging of dental charts, routine preventive care records, audit trail, and centralized storage of student dental records across schools with integration of a predictive analytics module to analyze historical dental records, identify potential dental health conditions, provide treatment recommendations, and risk classification interface through risk stratification categorization of dental health data to the dentist as inputs for a decision-support system. The study will be conducted within the academic year 2025 to 2026 and will encompass the software development process, from requirements analysis and system design to evaluation using the ISO 25010:2023 standard for software product quality.

Limitations

The study is limited to a web application and does not extend to mobile application development. The predictive analytics module is intended to function as a clinical decision support tool to assist in clinical decision-making and treatment planning, but does not replace the professional or clinical judgment of the dentist nor increase physical treatment capacity. The system will operate as a standalone platform without integration with national government databases or external health systems, and features such as computer vision-based caries detection, biometric authentication, and tele-dentistry are outside the scope of this study, as these functionalities are not part of the existing workflow and operational processes of the school dental clinics. Furthermore, the accuracy of the predictive module is dependent on the completeness and correctness of historical treatment data encoded into the system, and the system may require further clinical validation before it can be adapted for use in other districts or larger institutional settings.

Definition of Terms

Audit Trail. It refers to an automatic log recorded by the system that tracks all user actions such as additions, edits, and deletions made to student dental records and clinic data.

Dental Caries. It refers to tooth decay caused by bacterial acid, and is the primary dental health condition being monitored, recorded, and addressed through the proposed system.

Dental Charting. In this study, it refers to the digital odontogram within the system used by the dentist to record and update the condition of each tooth of a student during a clinic visit.

Dental Health. It refers to the overall condition of the teeth, gums, and surrounding oral structures, encompassing the prevention, diagnosis, and treatment of oral diseases and conditions. In this study, it refers specifically to the clinical condition of a student's teeth and gums as directly assessed and recorded by the school dentist during examination, including the presence or absence of dental caries, previous treatments received, and risk classification results used to guide preventive care planning, all of which are documented through oral examinations, dental charting, and treatment history recorded within the FLORAL system.

Dental Health Record Management System. In this study, it refers to the proposed web-based system that centralizes student dental records, appointment scheduling, treatment tracking, and predictive analytics across the school dental clinics of Barangay Tanyag.

Dental Health Risk Classification. In this study, it refers to the output generated by the predictive analytics module that categorizes each student into one of three risk levels (High, Medium, or Low) based on the analysis of historical dental records and patient health data, serving as the basis for treatment plan suggestions presented to the dentist for validation.

Decision Support System (DSS). In this study, it refers to the component of the system that presents treatment recommendations generated by the predictive analytics module to the dentist for review and validation before any clinical action is taken.

DMF/dmf Index. It refers to a standardized dental scoring system that quantifies a patient's history of tooth decay by counting the number of Decayed, Missing, and Filled teeth, where uppercase letters (DMF) are used for permanent dentition and lowercase letters (dmf) are used for primary or deciduous dentition, and is used in this study as a clinical input variable in the predictive analytics module.

DMFT Score. It refers to the standard measure used to quantify tooth decay in a population by counting a person's Decayed, Missing, and Filled Teeth, used in this study as a reference indicator of student dental health status.

Electronic Dental Records (EDR). It refers to the digital documentation of a patient's dental health information, used in this study to refer to existing digital dental health record systems whose findings support the development of the proposed system.

ISO 25010:2023. It refers to the international standard used in this study to evaluate the software quality of the proposed system across characteristics including functional suitability, usability, reliability, security, and performance efficiency.

National Objectives for Health (NOH). It refers to the DOH's medium-term health agenda that identifies dental health as a priority area and mandates the implementation of school-based dental health programs, providing the policy basis for this study.

Oral Health. It refers to the state of the mouth, teeth, gums, and related structures as defined by the Department of Health and World Health Organization. In this study, the term oral health is used in the context of DOH policy, official clinical documents, and public health mandates, consistent with the terminology used in the Individual Patient Treatment Record, the Oral Health Program Reporting Form, and the School Dental Health Care Program guidelines.

Oral Prophylaxis. It refers to the professional cleaning of the teeth performed by the school dentist or dental aide to remove plaque, debris, and calculus deposits, and is one of the components recorded under the Routine Preventive Care tracking module of the proposed system.

Predictive Analytics. In this study, it refers to the use of machine learning algorithms applied to historical student dental records to identify dental health risks, forecast possible dental conditions, and generate treatment recommendations.

Risk Stratification. In this study, it refers to the process of categorizing students according to their level of dental health risk based on their dental records and health data, enabling the dentist to prioritize care for those most in need.

Role-Based Access Control. In this study, it refers to the access management mechanism implemented in the proposed system that restricts each user's permissions and visible features based on their assigned role—specifically Dentist, Dental Aide, or School Administrator—ensuring that sensitive student health information is accessible only to authorized personnel.

Routine Preventive Care (RPC). In this study, it refers to the structured two-visit preventive dental care process administered to students, which includes dental screening, caries risk assessment, dental hygiene instruction, counseling, oral prophylaxis, and fluoride varnish application, in accordance with the School Dental Health Care Program guidelines.

Treatment Recommendation. In this study, it refers to the suggested dental intervention generated by the predictive analytics module based on a student's identified risk level, which the dentist reviews and validates before it is applied clinically

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